Lab 6: Modules, Interpolation and Texture Mapping

Computer Graphics Lab, Wolfram Mathematica

In[1]:=
```
SetOptions[EvaluationNotebook[], ShowCellLabel → True]
```

In[2]:=
```
SetOptions[{Plot, Plot3D, ContourPlot, ContourPlot3D, DensityPlot, ParametricPlot,
   ParametricPlot3D, ListPlot, ListLinePlot, ListPointPlot3D, Graphics, Graphics3D,
   PolarPlot, ListPlot}, ImageSize → Small, AxesLabel → {"x", "y", "z"}];
```

In[3]:=
```
(* Problem 1 *)
```

In[4]:=
```
f1[a_, b_] := Module[{s}, s = a + b; s]
```

In[5]:=
```
f1[5, 2]
```

Out[5]=
```
7
```

In[6]:=
```
(* Problem 2 *)
```

In[7]:=
```
f2[h1_, h2_, A_] := Module[{x}, x = h1[A] + h2[A] + A^2; x]
```

In[8]:=
```
h1[x_] := x^2
h2[x_] := Sin[x]
```

In[10]:=
```
f2[h1, h2, A]
```

Out[10]=
$$2\,A^2 + \text{Sin}[A]$$

In[11]:=
```
f2[h1, h2, Pi]
```

Out[11]=
$$2\,\pi^2$$

In[12]:=
```
(* Problem 3 *)
```

In[13]:=
```
f3[n_] := Module[{s}, s = Sum[i, {i, n}]; s]
```

In[14]:=
```
f3[5]
```

Out[14]=
```
15
```

In[15]:=
```
(* Problem 4 *)
```

In[16]:=
```
data4 = {1, 4, 5, 6, 8, 9, 1, 9, 7}
```

Out[16]=
```
{1, 4, 5, 6, 8, 9, 1, 9, 7}
```

In[17]:=
```
f4[data_] := Module[{f}, f := Interpolation[data, Method → "Spline"]; f]
```

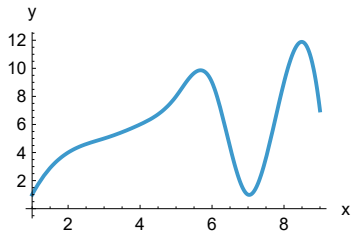
In[18]:=
```
myf = f4[data4]
```

Out[18]=

InterpolatingFunction [Domain: {{1, 9}} Output: scalar]

In[19]:=
```
Plot[myf[t], {t, 1, Length[data4]}]
```

Out[19]=



In[20]:=
```
(* Problem 5 *)
```

In[21]:=
```
data5 = {{1, 2}, {4, 4}, {6, 5}, {0, 0}, {2, 2}}
```

Out[21]=
```
{{1, 2}, {4, 4}, {6, 5}, {0, 0}, {2, 2}}
```

In[22]:=
```
f5[data_] := Module[{f, px, py, x, y},
  px = Transpose[data][[1]];
  py = Transpose[data][[2]];
  x = Interpolation[px, Method → "Spline"];
  y = Interpolation[py, Method → "Spline"];
  f = {x[#1], y[#1]} &;
  f]
```

In[23]:=
```
Off[InterpolatingFunction::dmval]
```

In[24]:=
```
Myf5 = f5[data5]
```

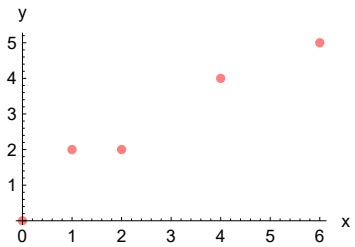
Out[24]=
```
{x$9079[#1], y$9079[#1]} &
```

In[25]:=
```
dd = Dimensions[data5][[1]]
```

Out[25]=
```
5
```

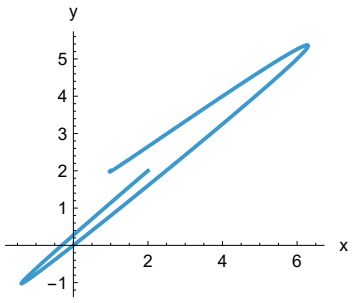
In[26]:=
```
pp = ListPlot[data5, PlotStyle → {Pink, PointSize[0.03]}]
```

Out[26]=



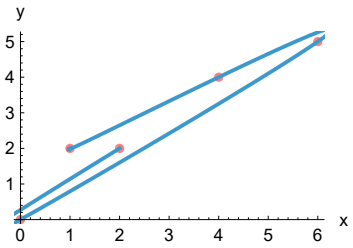
In[27]:=
```
p5 = ParametricPlot[f5[data5][t], {t, 1, dd}]
```

Out[27]=



In[28]:=
```
Show[pp, p5]
```

Out[28]=

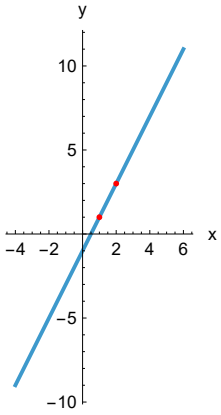


In[29]:=
```
(* Problem 6 *)
```

In[30]:=
```
GL2[s1_, s2_] := Module[{L, L1, G1, G2, GAll, t}, L[t_] := s1 * (1 - t) + s2 * t;
  L1 = ParametricPlot[L[t], {t, -5, 5}];
  G1 := Graphics[{Red, Point[s1]}];
  G2 := Graphics[{Red, Point[s2]}];
  GAll = Show[{L1, G1, G2}];
  GAll]
```

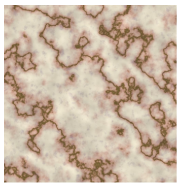
In[31]:=
```
GL2[{1, 1}, {2, 3}]
```

Out[31]=



In[32]:=
```
(* Problem 0 *)
```

In[33]:= **Text0 :=**

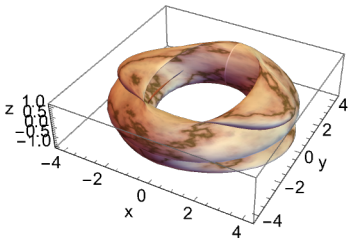


In[34]:=
```
a = 2.5;
```

In[35]:=
```
s0[u_, v_] := {(3 + Cos[a * v]) * Sin[u], (3 + Cos[v]) * Cos[u], Sin[v]}
```

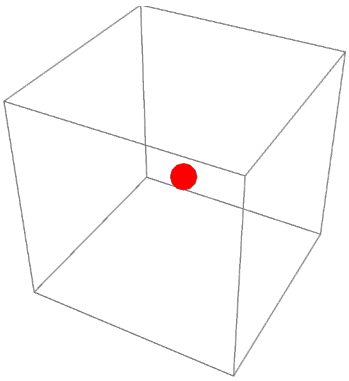
In[36]:=
```
G0 = ParametricPlot3D[s0[u, v], {u, 0, 2 * Pi},
   {v, 0, 2 * Pi}, PlotStyle → Texture[Text0], Mesh → False]
```

Out[36]=



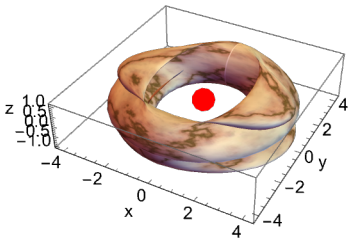
In[37]:=
```
Zero = Graphics3D[{Red, PointSize[0.08], Point[{{0, 0, 0}}]}]
```

Out[37]=



In[38]:=
```
Show[G0, Zero]
```

Out[38]=



In[39]:=
```
data0 := {{-4, 2, 4}, {4, 5, -1}, {2, 5, 0}, {1, 1, 3}, {2, -6, -6}, {4, 4, 4}}
xd0 = Transpose[data0][[1]];
yd0 = Transpose[data0][[2]];
zd0 = Transpose[data0][[3]];
sd0x = Interpolation[xd0, Method → "Spline"];
sd0y = Interpolation[yd0, Method → "Spline"];
sd0z = Interpolation[zd0, Method → "Spline"];
```

In[46]:=
```
Traj0[t_] = {sd0x[t], sd0y[t], sd0z[t]}
```

Out[46]=

$\{$InterpolatingFunction[ [t],

InterpolatingFunction[ [t],

InterpolatingFunction[ [t]$\}$

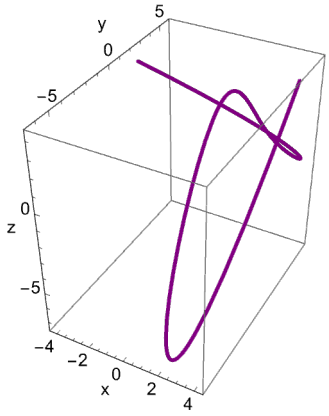
In[47]:=
```
L0 = Length[xd0];
```

In[48]:=
```
plot0t = ParametricPlot3D[Traj0[t], {t, 1, L0}, PlotStyle → Purple ];
```

In[49]:=
```
plot0tpoints = ListPointPlot3D[data0, PlotStyle → {PointSize[0.01], Blue}];
```

In[50]:=
```
GTraj0 = Show[{plot0t, plot0tpoints}]
```

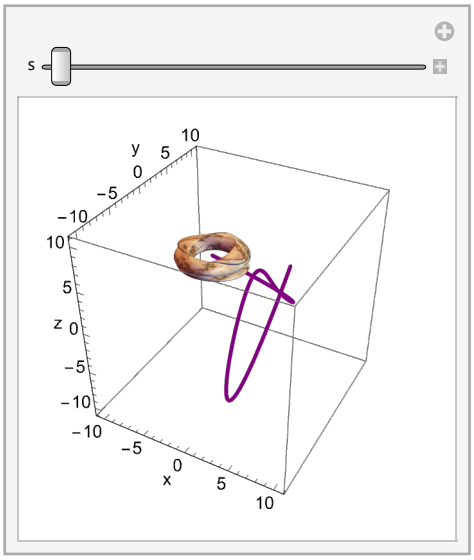
Out[50]=



In[51]:=
```
plots0[s_] := ParametricPlot3D[s0[u, v] + Traj0[s],
  {u, 0, 2 * Pi}, {v, 0, 2 * Pi}, PlotStyle → Texture[Text0], Mesh → False]
```

In[52]:=
```
Manipulate[Show[{GTraj0, plots0[s]}, PlotRange → 10], {s, 1, L0}]
```

Out[52]=



In[53]:=
```
Text01 := 
```

```
; Text11 := 
```

```
; Text12 := 
```


In[54]:=
```
Tex = {Text01, Text11, Text12}
```

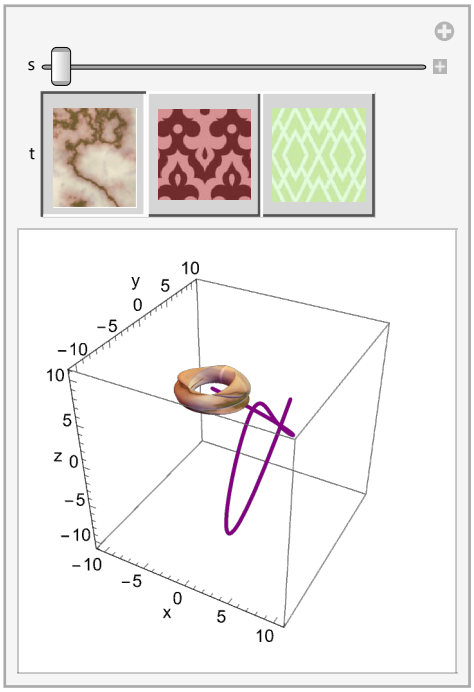
Out[54]=



In[55]:=
```
plotsT[s_, Tex_] := ParametricPlot3D[s0[u, v] + Traj0[s],
  {u, 0, 2 * Pi}, {v, 0, 2 * Pi}, PlotStyle → Texture[Tex], Mesh → False]
```

In[56]:=
```
Manipulate[Show[{GTraj0, plotsT[s, t]}, PlotRange → 10], {s, 1, L0}, {t, Tex}]
```

Out[56]=



In[57]:=
```
(* Problem (1) Interpolate a 3D data given by the heights of the explicit function z=
 f(x,y) using the synthetic data
   data1=Table[Round[Exp[x2-y^2],0.01]{x,-1,1,0.25},{y,-1,2,0.25}]
    Move the resulting surface along an interpolated curve
    through points pp1={0,0,0} and pp2={1,1,1} pp3={0.5,0.5,0.5}; pp4={0,3,2}
   ListInterpolation[data,{{xmin,xmax,{ymin,ymax}}]
*)
```

In[58]:=
```
pp1 = {0, 0, 0} ; pp2 = {1, 0, 0.5}; pp3 = {0.7, 0.8, 0.9}; pp4 = {0, 3, 2};
```

In[59]:=
```
fsyn[x_, y_] := x y Exp[x^2 - y^2]
```

In[60]:=
```
xmin1 = -1; xmax1 = 1; ymin1 = -1; ymax1 = 1;
```

In[61]:=
```
stepx1 = 0.25; stepy1 = stepx1;
```

In[62]:=
```
data1 =
  Table[Round[fsyn[x, y], 0.01], {x, xmin1, xmax1, stepx1}, {y, ymin1, ymax1, stepy1}];
```

In[63]:=
```
MatrixForm[data1]
```

Out[63]//MatrixForm=

1.	1.16	1.06	0.64	0.	−0.64	−1.06	−1.16	−1.
0.48	0.56	0.51	0.31	0.	−0.31	−0.51	−0.56	−0.48
0.24	0.27	0.25	0.15	0.	−0.15	−0.25	−0.27	−0.24
0.1	0.11	0.1	0.06	0.	−0.06	−0.1	−0.11	−0.1
0.	0.	0.	0.	0.	0.	0.	0.	0.
−0.1	−0.11	−0.1	−0.06	0.	0.06	0.1	0.11	0.1
−0.24	−0.27	−0.25	−0.15	0.	0.15	0.25	0.27	0.24
−0.48	−0.56	−0.51	−0.31	0.	0.31	0.51	0.56	0.48
−1.	−1.16	−1.06	−0.64	0.	0.64	1.06	1.16	1.

In[64]:=
```
data1
```

Out[64]=
```
{{1., 1.16, 1.06, 0.64, 0., -0.64, -1.06, -1.16, -1.},
 {0.48, 0.56, 0.51, 0.31, 0., -0.31, -0.51, -0.56, -0.48},
 {0.24, 0.27, 0.25, 0.15, 0., -0.15, -0.25, -0.27, -0.24},
 {0.1, 0.11, 0.1, 0.06, 0., -0.06, -0.1, -0.11, -0.1},
 {0., 0., 0., 0., 0., 0., 0., 0., 0.}, {-0.1, -0.11, -0.1, -0.06, 0., 0.06, 0.1, 0.11, 0.1},
 {-0.24, -0.27, -0.25, -0.15, 0., 0.15, 0.25, 0.27, 0.24},
 {-0.48, -0.56, -0.51, -0.31, 0., 0.31, 0.51, 0.56, 0.48},
 {-1., -1.16, -1.06, -0.64, 0., 0.64, 1.06, 1.16, 1.}}
```

In[65]:=
```
DR1
```

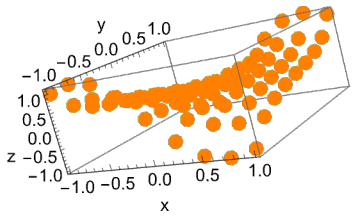
Out[65]=
```
DR1
```

In[66]:=
```
(* x-y-range *)
```

In[67]:=
```
DR1 = {{xmin1, xmax1}, {ymin1, ymax1}};
```

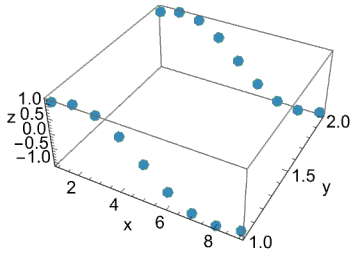
In[68]:=
```
GD1 = ListPointPlot3D[data1, PlotStyle → {Orange, PointSize[0.05]}, DataRange → DR1]
```

Out[68]=



In[69]:=
```
ListPointPlot3D[{{1., 1.16, 1.06, 0.64, 0., -0.64, -1.06, -1.16, -1.},
  {1., 1.16, 1.06, 0.64, 0., -0.64, -1.06, -1.16, -1.}}]
```

Out[69]=



In[70]:=
```
(* Transpose the matrix to comply with ListInterpolation. *)
```

In[71]:=
```
Tdata1 = Transpose[data1];
```

In[72]:=
```
f1 = ListInterpolation[Tdata1, DR1]
```

Out[72]=

InterpolatingFunction[Domain: {{−1., 1.}, {−1., 1.}} Output: scalar]

In[73]:=
```
Zero = Graphics3D[{Red, PointSize[0.08], Point[{{0, 0, 0}}]}];
```

In[74]:=
```
(* Plot the surface *)
```

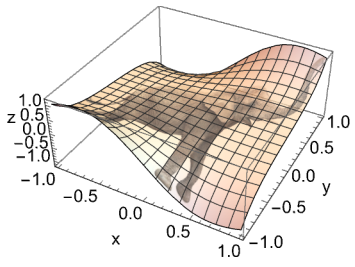
In[75]:=
```
TextEl =
```



```
;
```

In[76]:=
```
Ginter = Plot3D[f1[x, y], {x, xmin1, xmax1},
  {y, ymin1, ymax1}, PlotStyle → {Blue, Opacity[0.7], Texture[TextEl]}]
```

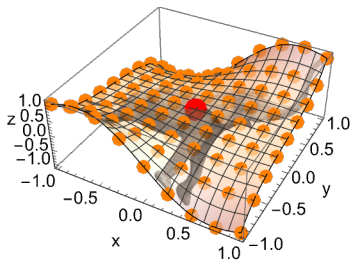
Out[76]=



In[77]:=
```
(* Superimpose  *)
```

In[78]:=
```
GAll = Show[{GD1, Ginter, Zero}, Boxed → True]
```

Out[78]=



In[79]:=
```
(* Understand the structure of the modules below  *)
```

In[80]:=
```
Listp = 2 {pp1, pp2, pp3, pp4}
```

Out[80]=
```
{{0, 0, 0}, {2, 0, 1.}, {1.4, 1.6, 1.8}, {0, 6, 4}}
```

In[81]:=
```
(* Extract x,y,z from the data  *)
```

In[82]:=
```
Extractxyz[data_] := Module[{dataT, px, py, pz},
  dataT = Transpose[data];
  px = dataT[[1]];
  py = dataT[[2]];
  pz = dataT[[3]];
  {px, py, pz}(*output=x,y,z*)]
```

In[83]:=
```
{px, py, pz} = Extractxyz[Listp]
```

Out[83]=
```
{{0, 2, 1.4, 0}, {0, 0, 1.6, 6}, {0, 1., 1.8, 4}}
```

In[84]:=
```
(* Generate fx,fy,fz to compose a parametric function Traj1[s]  fill 4 lines here*)
```

In[85]:=
```
Interpxyz[data_] := Module[{px, py, pz, fx, fy, fz},
  {px, py, pz} = Extractxyz[data];
  fx = Interpolation[px, Method → "Spline"];
  fy = Interpolation[py, Method → "Spline"];
  fz = Interpolation[pz, Method → "Spline"];
  {fx, fy, fz}]
```

In[86]:=
```
{tx, ty, tz} = Interpxyz[Listp]
```

Out[86]=

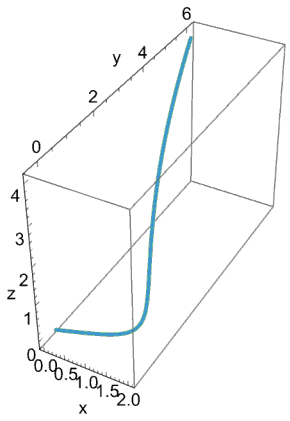
{InterpolatingFunction[Domain: {{1., 4.}} Output: scalar],

InterpolatingFunction[Domain: {{1., 4.}} Output: scalar], InterpolatingFunction[Domain: {{1., 4.}} Output: scalar]}

In[87]:=
```
Traj1[s_] := {tx[s], ty[s], tz[s]}
```

In[88]:=
```
GL1 = ParametricPlot3D[Traj1[t], {t, 1, 4}]
```

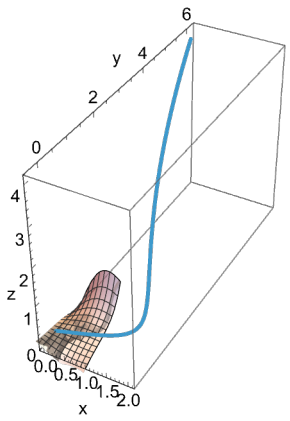
Out[88]=



In[89]:=
```
(* The line and the surface  *)
```

In[90]:=
```
SG11 = Show[GL1, Ginter]
```

Out[90]=



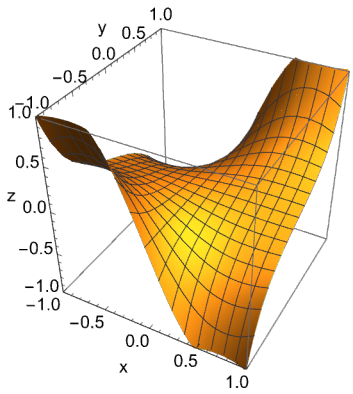
In[91]:=
```
(* Convert the surface into the parametric form  *)
```

In[92]:=
```
S1[u_, v_] := {u, v, f1[u, v]}
```

In[93]:=
```
(* Check  *)
```

In[94]:=
```
GP1 = ParametricPlot3D[S1[x, y],
  {x, xmin1, xmax1}, {y, ymin1, ymax1}, PlotRange → 1, BoxRatios → 1]
```

Out[94]=



In[95]:=
```
SurfacePosition1[u_, v_, t_] := S1[u, v] + Traj1[t]
```

In[96]:=
```
(* Check  *)
```

In[97]:=
```
SurfacePosition1[1, 1, 1]
```

Out[97]=
```
{1., 1., 1.}
```

In[98]:=
```
Text1 :=
```

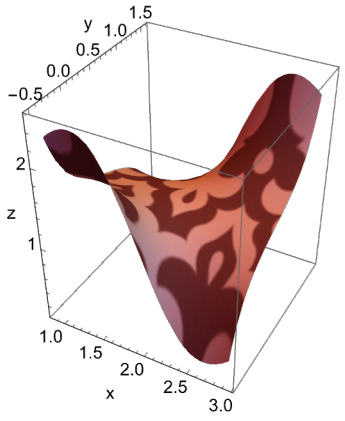

In[99]:=
```
Gfunction1[t_] := ParametricPlot3D[SurfacePosition1[x, y, t],
  {x, xmin1, xmax1}, {y, ymin1, ymax1}, PlotStyle → Texture[Text1], Mesh → False]
```

In[100]:=
```
(* Check  *)
```

In[101]:=

```
Gfunction1[2.5]
```

Out[101]=



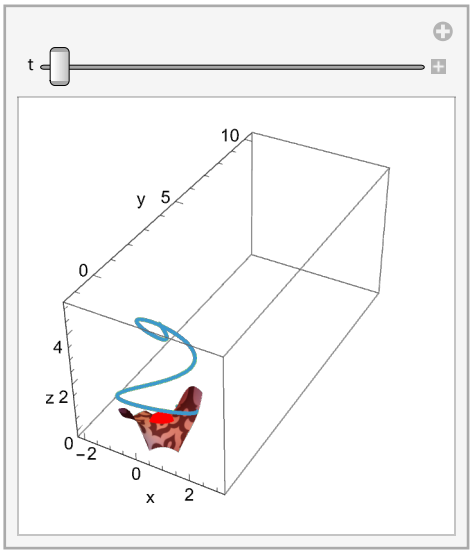
In[102]:=

```
GShowAll[s_] := Show[{GL1, Zero, Gfunction1[s]}, PlotRange → {{-2, 3}, {-1, 10}, {0, 5}}]
```

In[103]:=

```
Manipulate[GShowAll[t], {t, 1, 4}]
```

Out[103]=



In[104]:=

```
(* Problem 2 *)
```

In[105]:=

```
data0 = Table[{Sin[3 * t], Cos[2 * t], t}, {t, 0, 5, 0.25}]
```

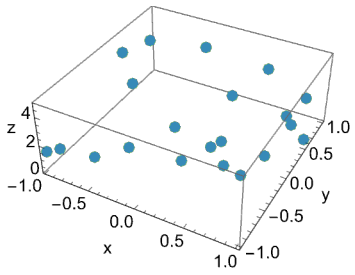
Out[105]=

```
{{0., 1., 0.}, {0.681639, 0.877583, 0.25}, {0.997495, 0.540302, 0.5},
 {0.778073, 0.0707372, 0.75}, {0.14112, -0.416147, 1.}, {-0.571561, -0.801144, 1.25},
 {-0.97753, -0.989992, 1.5}, {-0.858934, -0.936457, 1.75}, {-0.279415, -0.653644, 2.},
 {0.450044, -0.210796, 2.25}, {0.938, 0.283662, 2.5}, {0.922604, 0.70867, 2.75},
 {0.412118, 0.96017, 3.}, {-0.319519, 0.976588, 3.25}, {-0.879696, 0.753902, 3.5},
 {-0.967808, 0.346635, 3.75}, {-0.536573, -0.1455, 4.}, {0.182599, -0.602012, 4.25},
 {0.803784, -0.91113, 4.5}, {0.993641, -0.997172, 4.75}, {0.650288, -0.839072, 5.}}
```

In[106]:=

```
ppp = ListPointPlot3D[data0]
```

Out[106]=



In[107]:=

```
Extractxyz[data0]
```

Out[107]=

```
{{0., 0.681639, 0.997495, 0.778073, 0.14112, -0.571561, -0.97753,
  -0.858934, -0.279415, 0.450044, 0.938, 0.922604, 0.412118, -0.319519,
  -0.879696, -0.967808, -0.536573, 0.182599, 0.803784, 0.993641, 0.650288},
 {1., 0.877583, 0.540302, 0.0707372, -0.416147, -0.801144, -0.989992, -0.936457,
  -0.653644, -0.210796, 0.283662, 0.70867, 0.96017, 0.976588, 0.753902, 0.346635,
  -0.1455, -0.602012, -0.91113, -0.997172, -0.839072}, {0., 0.25, 0.5, 0.75, 1.,
  1.25, 1.5, 1.75, 2., 2.25, 2.5, 2.75, 3., 3.25, 3.5, 3.75, 4., 4.25, 4.5, 4.75, 5.}}
```

In[108]:=

```
{tx1, ty1, tz1} = Interpxyz[data0]
```

Out[108]=

```
{InterpolatingFunction[ Domain: {{1., 21.}}
                        Output: scalar ],

 InterpolatingFunction[ Domain: {{1., 21.}}
                        Output: scalar ], InterpolatingFunction[ Domain: {{1., 21.}}
                                                                 Output: scalar ]}
```

In[109]:=

```
Length[data0]
```

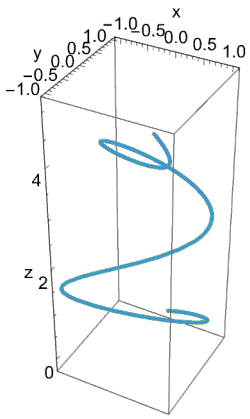
Out[109]=

```
21
```

In[110]:=

```
GL1 = ParametricPlot3D[{tx1[t], ty1[t], tz1[t]}, {t, 1, Length[data0]}]
```

Out[110]=



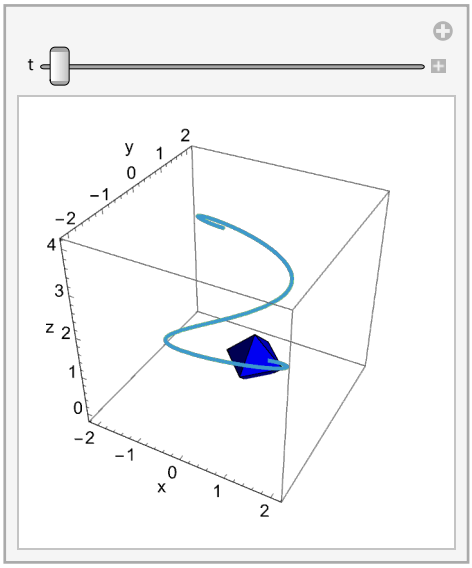
In[111]:=

```
pp5[t_] := Graphics3D[{Blue, Octahedron[{tx1[t], ty1[t], tz1[t]}, 1]}]
```

In[112]:=

```
Manipulate[Show[{GL1, pp5[t]}, PlotRange → {{-2, 2}, {-2, 2}, {0, 4}}],
 {t, 1, Length[data0]}]
```

Out[112]=



In[113]:=

```
(* Bonus *)
```

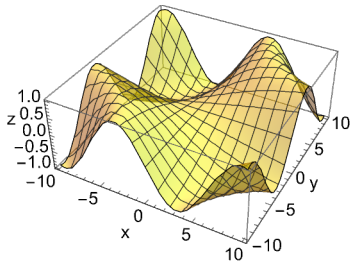
In[114]:=

```
S[x_, y_] := Sin[x * y / 20]
```

In[115]:=

```
s1 = Plot3D[S[x, y], {x, -10, 10}, {y, -10, 10}, PlotStyle → {Yellow, Opacity[0.5]}]
```

Out[115]=



In[116]:=

```
DR1
```

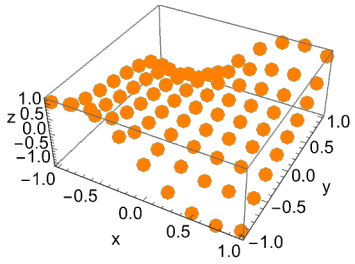
Out[116]=

```
{{-1, 1}, {-1, 1}}
```

In[117]:=

```
GD1 = ListPointPlot3D[data1, PlotStyle → {Orange, PointSize[0.05]}, DataRange → DR1]
```

Out[117]=



In[118]:=

```
data1 = Table[Round[fsyn[x, y], 0.01], {x, xmin1, xmax1, stepx1}, {y, ymin1, ymax1, stepy1}];
```

In[119]:=

```
data1
```

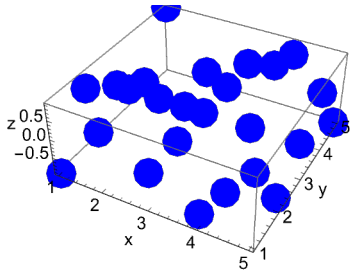
Out[119]=

```
{{1., 1.16, 1.06, 0.64, 0., -0.64, -1.06, -1.16, -1.},
 {0.48, 0.56, 0.51, 0.31, 0., -0.31, -0.51, -0.56, -0.48},
 {0.24, 0.27, 0.25, 0.15, 0., -0.15, -0.25, -0.27, -0.24},
 {0.1, 0.11, 0.1, 0.06, 0., -0.06, -0.1, -0.11, -0.1},
 {0., 0., 0., 0., 0., 0., 0., 0., 0.}, {-0.1, -0.11, -0.1, -0.06, 0., 0.06, 0.1, 0.11, 0.1},
 {-0.24, -0.27, -0.25, -0.15, 0., 0.15, 0.25, 0.27, 0.24},
 {-0.48, -0.56, -0.51, -0.31, 0., 0.31, 0.51, 0.56, 0.48},
 {-1., -1.16, -1.06, -0.64, 0., 0.64, 1.06, 1.16, 1.}}
```

In[120]:=

```
data = Table[S[x, y], {x, -10, 10, 5}, {y, -10, 10, 5}];
s2 = ListPointPlot3D[data, PlotStyle → {Blue, PointSize[0.1]}]
```

Out[121]=



In[122]:=

```
PlotSpheres[S_, N_, RS_] := Module[{sur, step, points, pointsf, sphereAll, result, p},
  sur = Plot3D[S[x, y], {x, -10, 10}, {y, -10, 10}, PlotStyle → {Yellow, Opacity[0.5]}];
  step = 20 / (N - 1);
  p[i_] := -10 + step * (i - 1);
  points = Table[{p[i], p[j], S[p[i], p[j]] + RS}, {i, N}, {j, N}];
  pointsf = Flatten[points, 1];
  sphereAll = Graphics3D[{Blue, Sphere[pointsf, RS]}];
  result = Show[{sphereAll, sur}, PlotRange → Automatic];
  result]
```

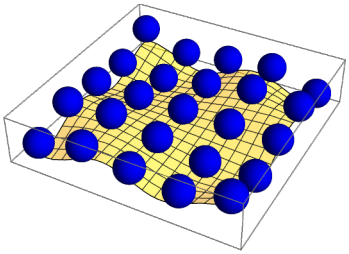
In[123]:=

```
S[x_, y_] := Sin[x * y / 20]
```

In[124]:=

```
PlotSpheres[S, 5, 1.5]
```

Out[124]=



In[125]:=

```
PlotSpheres1[S_, N_, RS_] := Module[{s1, s2, data, res},
  s1 = Plot3D[S[x, y], {x, -10, 10}, {y, -10, 10}, PlotStyle → {Yellow, Opacity[0.5]}];
  data = Table[S[x, y], {x, -10, 10, 20 / (N - 1)}, {y, -10, 10, 20 / (N - 1)}];
  s2 = ListPointPlot3D[data, PlotStyle → {Blue, PointSize[RS]}];
  res = Show[{s1, s2}];
  res]
```

In[126]:=

```
PlotSpheres1[S, 5, 0.1]
```

Out[126]=

